The terminal singleton supplies an opposite head

Human Compression project: L044 continuation

Retain the complete critical six-point-head configuration and its acquired proofs. Thus $G \leq \mathrm{Alt}(\Omega)$ is transitive with connected string C-marking, all maximal omissions are intransitive, and $n = 2r \geq 18$. The first three marks are an actual supported extendable six-point head on D, with exit vertex v and

$$Q = G_{\geq 3} = \mathrm{Alt}(W), \qquad W = \Omega \setminus (D \setminus \{v\}).$$

Assume the terminal omission is the actual natural group

$$P = G_{\leq r-2} = \mathrm{Alt}(\Omega \setminus \{z\}), \tag{1}$$

and every colour $3, \ldots, r-2$ has two full ambient-eligible edges. This last hypothesis is explicit; it holds when the terminal colour is the first unique colour after the head, after the acquired exclusions of colours $3, 4$. The induction that produced the critical setting remains inherited. No new rank induction is used below. Put

$$\eta = \rho_{r-1}, \quad \sigma = \rho_{r-2}, \quad H = G_{\leq r-3}, \quad K = G_{\leq r-4}, \quad a = z^{\eta}.$$

Transitivity gives $a \neq z$. Since η centralizes H and H fixes z, it also fixes a. Thus $\eta = (z\ a)\epsilon$, where ϵ is an odd involution fixing z, a and centralizing H. The terminal mark centralizes the actual head, so fixes D pointwise. Hence z, a and the support of ϵ lie outside D.

Lemma 1 (The actual last mark is a double transposition)**.** *There are distinct points u, v' outside $D \cup \{z, a\}$ such that*

$$\eta = (z\ a)(u\ v'),$$

and K fixes u, v' pointwise. The group H preserves their pair.

Proof. Apply the complete odd-singleton theorem to Q on W, with the entire marked string reversed. Its rank is $p = r - 3 \geq 6$ and its degree is $n - 5 = 2p + 1$. The actual first omission is

$$Q \cap P = \mathrm{Alt}(W \setminus \{z\}),$$

by (1) and full ambient IP. Restriction to W is faithful, so IP, nonidentity marks, rank and connectedness are retained. Every maximal omission is intransitive: a transitive omission on W, together with the left head meeting W only at v, would make its corresponding ambient omission transitive. The required reversed colours at least two are precisely the original colours $3, \ldots, r-3$. Their ambient-eligible edges lie in W and remain eligible for the smaller Q-omissions. This verifies every hypothesis of that theorem.

Write $H_Q = \langle \rho_3, \ldots, \rho_{r-3} \rangle$ and $K_Q = \langle \rho_3, \ldots, \rho_{r-4} \rangle$. The theorem gives either at least three H_Q-fixed points in W, or the literal residual $\eta = (z\ a)(u\ v')$, whose two-point orbit $E = \{u, v'\}$ is fixed pointwise by K_Q. The latter fixation is proved in the theorem's complete second proof, including its localized contradiction when K_Q moves that pair. Since E is disjoint from D, the first three marks fix it as well, so K fixes it. This establishes the lemma in the literal residual.

Suppose instead that H_Q has at least three fixed points. None is v, because its literal first original mark ρ_3 moves v. The left head fixes $W \setminus \{v\}$, so these are also fixed by H. Put $U = \mathrm{Supp}(H)$, $F = \Omega \setminus U$ and $d = r - 2$, so $|F| \geq 3$. The pure-prefix theorem gives a head orbit A of H carrying pure $\mathrm{Alt}(A)$. Its exact intersection with Q removes five head points and

leaves all exterior orbits unchanged. If q counts the nontrivial H-orbits, the paired interval $H \cap Q$ of rank $d-3$ gives

$$2(d-3) \le |U| - 5 - q, \qquad |U| \ge 2d - 1 + q. \tag{2}$$

As $|U| \le n - 3 = 2d + 1$, this forces $q \le 2$ and $|F| \le 4$.

The involution ϵ centralizes H and fixes A pointwise, since H contains pure $\mathrm{Alt}(A)$. For $q = 1$, it therefore fixes all of U. For $q = 2$, (2) forces $|U| = 2d + 1$. Let E_0 be the other nontrivial orbit. If ϵ acted nontrivially on E_0, it would be free there. Partition the $d-3$ paired colours of $H \cap Q$ into those with an eligible edge in E_0 and the rest. The first set has at most $|E_0|/2 - 1$ colours: project one eligible edge per colour to the ϵ-pairs; the actual η belongs to every relevant ambient omission and supplies the pair switches, so an alternate projected path would contradict eligibility. The projected graph is a forest. All eligible edges of the second set lie in the single orbit $A \cap W$, so the complete selected-localization theorem bounds that set by $(|A| - 6)/2$. Their total is at most

$$\frac{|E_0|}{2} - 1 + \frac{|A| - 6}{2} = \frac{2d + 1 - 8}{2} = d - \frac{7}{2} < d - 3,$$

a contradiction. Thus ϵ fixes U in both cases.

The at most four points of F include the fixed points z, a. Oddness makes ϵ precisely one transposition on the remaining points. Its pair E is fixed pointwise by H, and hence by K. Finally in either branch, H centralizes ϵ and fixes z, a, so preserves E. □

Lemma 2 (The actual penultimate mark is a double transposition)**.** *Under the same hypotheses, σ fixes all but four points.*

Proof. Write $E = \{u, v'\}$ as in Lemma 1, and put $x = a^\sigma$. As $P = \langle H, \sigma \rangle$ is transitive on $\Omega \setminus \{z\}$ and H fixes a, we have $x \ne a$. Also σ fixes z. The group K centralizes σ, so fixes pointwise

$$z, a, E, E^\sigma, x. \tag{3}$$

It has at most six fixed points. Indeed its rank is $r - 3$, and its pure-prefix intersection with Q removes five head points while retaining its $q_K \ge 1$ nontrivial orbits. Every colour of $K \cap Q = G_{\{3,\dots,r-4\}}$ is paired, so the complete ambient interval inequality gives

$$2(r-6) \le |\operatorname{Supp}(K)| - 5 - q_K, \qquad |\operatorname{Supp}(K)| \ge 2r - 6. \tag{4}$$

Thus (3) cannot contain seven distinct points.

If $x \in E$, name it u. Then $u^\sigma = a$ and $w = (v')^\sigma$ lies outside $\{z, a, u, v'\}$. Injectivity and fixation of z exclude the first three values; $w = v'$ would make $\{a, u, v'\}$ invariant under P, since H fixes a and preserves E. Thus the core of σ is $(a\ u)(v'\ w)$ on five points, including the fixed z.

If $x \notin E$, the set E^σ cannot equal E, since P would preserve E. If it were disjoint from E, then z, a, x, E, E^σ would be seven distinct fixed points of K, contrary to (4). Their intersection therefore has one point. As σ is an involution, that point is fixed by it; name it v'. Then $u^\sigma = w \notin \{z, a, x, u, v'\}$, and the core is $(a\ x)(u\ w)$ on six points, including the fixed z, v'.

In either case write σ as the displayed two transpositions times a disjoint involution ν. The latter is even, since σ is even. The product $\eta\sigma$ on the core has order five in the first case and order three in the second; in the second case it consists of two disjoint three-cycles. Therefore $\nu = (\eta\sigma)^5$ or $(\eta\sigma)^3$, respectively. It fixes z, so by (1) it belongs to P. Full IP gives

$$\nu \in \langle \eta, \sigma \rangle \cap P = \langle \sigma \rangle.$$

But ν fixes a and σ moves a. Thus $\nu = 1$, and σ is literally the displayed double transposition. □

Theorem 3. *The actual last three marks form the pure six-point A_5 marking of source Table 2(3), read from the terminal end.*

Proof. Apply the complete terminal-double-transposition theorem. Its natural alternating prefix is exactly (1). The group $H = G_{\le r-3}$ has its actual head orbit of size at least four and contains its pure alternating group by the acquired prefix-kernel theorem. Lemma 2 supplies the literal double transposition ρ_{r-2}. These are every hypothesis of that attachment theorem, which gives the asserted three actual marks and removes all exterior decoration through its full IP argument. □

Attribution and boundary. This supplies the terminal-singleton attachment in the route of Proposition 5.32 of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1. The complete odd-singleton theorem, pure-prefix kernels, ambient localization and terminal attachment are acquired. The present proof verifies the actual odd-tail embedding, transfers its short fixed set past the supported left head, and extracts the two literal final double transpositions. It retains the full paired-interior hypothesis. Its conclusion is an actual opposite head; excluding that branch and completing all source routing are separate tasks. No whole-original, question, numerical or formal credit is claimed.